TraceSOC Technical Validation and Evidence Reconstruction Report

Architecture, engineering progression, evidence reconstruction, independent proof, limitations and defense

> TraceSOC is a provenance-first SOC/DFIR evidence-reconstruction engine designed so that consequential downstream claims remain mechanically traceable to exact raw evidence while observation, inference, uncertainty and response boundaries remain explicit.

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Project / variant: SOC-A5 / V3
Candidate: UBI-2026-0091
Evidence marker: UBI-A9-4BC183701F3C
Frozen technical commit: 8751a679f36b4983cdd3d2b0b35f44e9a11c4f39
Frozen technical tree: fc420ee3fd1988cf578c0cf309ee3aefcccf33b2

Submission note: This report is evidence-bounded. It distinguishes measured capability from unmeasured accuracy, causality, maliciousness, live response execution, and ROI.

Contents and reading path

The report is structured to answer UBI's four finalist questions first, then show the complete engineering and proof story behind those answers.

Section	Purpose
1. Problem, Impact, Proof, Defense	Direct answers to the four finalist questions.
2. Project origin and engineering progression	How lessons became a reusable frozen product rather than a case-specific solution.
3. Product objective and evidence contract	Forward and reverse analytical requirements.
4. Supported evidence boundary	Frozen representations, explicit non-goals and no-silent-loss behavior.
5. Provenance, identity and observation discipline	Canonicalization and separation of fact from inference.
6. Detection and bounded correlation	Six correlation families, selectors and claim boundaries.
7. Controlled six-source reconstruction	R3 measured state and the full supported workflow.
8. Packet accounting and aggregation repair	How evidence multiplicity was preserved without inflating conclusions.
9. End-to-end reverse trace	Mechanical downstream-to-original-SHA verification.
10. External evidence and holdout	Generalization controls, freeze discipline and unadjudicated accuracy.
11. Regression, anti-hardcoding and reproduction	Software assurance and deterministic accepted outputs.
12. Resource validation	Measured viability on the constrained workstation.
13. Operational and business significance	Value supported by the evidence, without invented ROI.
14. Limitations, trade-offs and future validation	What remains unresolved and why.
15. Judge verification procedure and evidence map	Exact independent verification path and selected authority hashes.
16. Defense guide	Major technical choices and likely challenge questions.
17. Conclusion	Bounded technical claim.

Problem, Impact, Proof, Defense

BEGIN WITH THESE QUESTIONS

Problem, Impact, Proof, Defense

UBI asks four questions at the start of the finalist story. TraceSOC can answer all four directly, and each answer is tied to measured evidence rather than marketing claims.

Problem - What real security problem are you solving, and for whom?

SOC analysts and DFIR investigators work across fragmented email, endpoint, packet and network evidence. As investigations progress, conclusions can become detached from the exact observations that support them. That weakens reproducibility, analyst handoff, auditability and later technical review. TraceSOC is for SOC analysts, incident responders, DFIR practitioners, and the reviewers or security teams that need to validate how a conclusion was reached.

Impact - What becomes safer, faster, clearer, or more resilient because your project exists?

TraceSOC makes investigations clearer and more defensible, safer from unsupported analytical certainty, and more resilient to handoff and later review by preserving exact evidence lineage and explicit uncertainty. Deterministic reproduction improves repeatability. Analyst-time reduction was not measured, so no claim that investigations are faster is made.

Proof - How will a judge independently verify that it works?

The judge can verify the pre-holdout technical freeze; 5,442/5,442 raw-locator coverage in the controlled six-source case; one complete response-to-original-SHA reverse trace; 23,145/23,145 external Zeek reconciliation; 156,458/156,458 sealed-holdout reconciliation with 938,748 detector evaluations and no post-freeze tuning; 524/524 regression with 4/4 public acceptance; two fresh reproductions with the same 11 core hashes; and the measured 198,463-record resource benchmark.

Defense - Can you explain every major technical choice, limitation, and trade-off?

Yes. The architecture deliberately treats provenance as part of the event contract, preserves unsupported evidence, separates observation from inference, uses representation-specific adapters, bounds correlation to what evidence supports, aggregates downstream reasoning without deleting packet multiplicity, freezes before holdout evaluation, distinguishes recommendations from executed response, and records open limitations instead of inventing accuracy, causality, maliciousness, ROI or cross-publisher generalization.

From Stage 9A lessons to a reusable evidence-reconstruction engine

TraceSOC was not designed to reproduce a previous case answer. Stage 9A is continuity and compatibility evidence: it exposed transferable engineering lessons, but it does not define the finalist product.

What was carried forward

- Raw evidence provenance as a non-negotiable contract.
- Explicit separation between observed facts and downstream inference.
- Deterministic identity rather than fixture-dependent identity.
- Bounded correlations that carry alternatives and falsifiers.
- Clean reproducibility as part of technical acceptance.
- Anti-hardcoding and pre-freeze holdout discipline.
- Safe response semantics that distinguish recommendation/simulation from executed action.

What was deliberately not carried forward

- Case-specific Stage 9A answers.
- Prior production runtime as TraceSOC runtime.
- Historical test denominators as current finalist regression truth.
- Any assumption that one previously known evidence set defines the supported parser boundary.

> The product objective became broader and stricter: reconstruct supported raw evidence into defensible analytical objects while preserving a reversible chain back to exact source evidence.

Engineering progression

1. Ground the product in Stage 9B/finalist requirements, reusable provenance contracts and transferable grader lessons.
2. Define explicit supported representations and preserve unsupported/partial evidence rather than silently dropping it.
3. Build and validate canonical observations and semantic events across real evidence families.
4. Add detectors and bounded cross-source correlations with evidence-specific selectors.
5. Demonstrate a controlled six-source vertical slice, then repair analytical inflation without deleting packet evidence.
6. Prove complete reverse traceability from downstream output to original artifact SHA-256.
7. Challenge overfitting with anti-hardcoding checks and external real evidence.
8. Freeze the technical candidate before semantic holdout access.
9. Adjudicate the sealed holdout without post-freeze tuning.
10. Prove current regression, public acceptance, deterministic reproduction and measured resource behavior.

The evidence contract: forward analysis and reverse proof

TraceSOC is not primarily a dashboard, SIEM replacement or alert generator. It is an evidence-reconstruction engine whose central contract is that analytical convenience must not destroy evidence identity.

Forward analytical path

```
raw evidence
-> representation extraction
-> canonical observation
-> evidence graph
-> detection
-> relationship / correlation
-> hypothesis / finding
-> narrative
-> response
```

Reverse evidentiary requirement

```
downstream conclusion
-> finding
-> relationship / detection
-> canonical observation
-> exact raw locator
-> original artifact
-> SHA-256
```

Core architectural principles

Principle	Why it exists
Provenance-first	A downstream conclusion must be recoverable to the exact raw source that supports it.
Observation before inference	The system must not blur what the artifact said with what an analyst or rule inferred.
Deterministic identity	Object identity must not depend on fixture IDs, expected answers or verdict labels.
Bounded correlation	Relationships must be no stronger than the selectors and evidence actually support.
No silent evidence loss	Unsupported or partial records must remain accounted rather than disappear.
Reproducibility	Accepted outputs should be regenerated by the frozen product from accepted inputs.
Explicit response state	A recommended or simulated response must not be presented as executed containment.

Supported evidence families, frozen representations and explicit non-goals

The controlled reference demonstrates six evidence families. The frozen product boundary is expressed at the representation level, yielding twelve supported forms across those families.

Evidence family	Frozen supported representations
Email	EML / MIME
Packet	PCAP; PCAPNG
Zeek	Standard ASCII/header-aware conn.log; native JSON/JSONL
Sysmon	Windows Event XML; recognized native Windows Event JSON/JSONL
PowerShell	Operational Windows Event XML; recognized native Windows Event JSON/JSONL
Windows Security	Windows Event XML; recognized native Windows Event JSON/JSONL; supported Splunk-style text export

Explicitly out of scope

- MBOX and MSG email containers.
- Native EVTX ingestion.
- Arbitrary SIEM/vendor JSON or CSV schemas.
- Universal log parsing or undocumented packet/Zeek formats.
- Heavyweight SIEM infrastructure as a deployment prerequisite.
- Destructive or live production response execution.

> Out of scope does not mean discarded. Unknown/unsupported representations are preserved/accounted so the system does not create false completeness.

Preserving exact evidence while separating observation from inference

Canonical observations as evidence-bearing objects

The canonical layer exists to create consistent analytical objects without severing raw provenance. In accepted R3, all 5,442 canonical observations carry raw locators. That allows downstream proof to resolve back to source evidence rather than stop at a normalized summary.

Observation and inference remain different layers

Layer	What it is allowed to claim	What it must preserve
Raw/canonical observation	What the source actually contains.	Source identity, raw locator, artifact identity, representation context.
Detection	A rule matched a semantic condition.	Contributing observation/event and rule identity.
Correlation	Evidence-backed relationship between supported objects.	Selectors, corroboration basis, alternatives/falsifiers, claim boundary.
Finding	Analytical conclusion derived from evidence and relationships.	Supporting objects, confidence, alternative interpretation, residual uncertainty.
Narrative	Human-readable explanation of the evidence-backed finding.	Traceability to findings rather than free-standing claims.
Response	Recommendation or validation state.	Explicit executed/not-executed distinction.

Why this matters

Without this separation, a system can accidentally promote a correlation into causality, a controlled value into a malicious IOC, or a recommendation into proof of containment. TraceSOC treats those as category errors and keeps them visible in the data model and reporting layer.

Evidence-supported relationships without unsupported causality

TraceSOC implements a bounded set of correlation families. Each rule exists because a specific evidence-supported relationship can be established; no rule is permitted to silently add maliciousness, user intent or causal direction that the source data does not prove.

ID	Relationship	Primary selector / support	Does not prove
TRC-CORR-001	PowerShell web retrieval <-> Zeek	Supported network/retrieval relationship	Causality or maliciousness
TRC-CORR-002	PowerShell <-> Sysmon process	Process/evidence corroboration	Compromise verdict
TRC-CORR-003	Sysmon process <-> PCAP network	Supported process/network relationship	Intent or maliciousness
TRC-CORR-004	Windows Security 4688 <-> Sysmon Event 1	Same host, PID, exact image; time bounded but never sole selector	Maliciousness or causality
TRC-CORR-005	EML attachment <-> Sysmon process image	Exact SHA-256 byte identity	Delivery, click or execution causality
TRC-CORR-006	Sysmon Event 1 process <-> Event 3 network	Exact normalized ProcessGuid; host consistency	Maliciousness; network-before-process is rejected

Why time is not enough

Close timestamps are useful corroboration but are not a safe sole identity selector. The design therefore treats time as bounded corroborative context rather than a substitute for stronger identifiers such as exact image, host/PID combinations, SHA-256 identity or ProcessGuid.

Two genuine pre-freeze defects found and fixed during CORR-006

- The correlation record whitelist omitted sysmon_network_activity.
- The source-kind whitelist omitted sysmon_process.

These were technical defects in the reusable correlation path and were corrected before the finalist freeze. They were not post-holdout tuning.

R3: one coherent workflow across six evidence families

The controlled reference case was designed as interoperability and provenance evidence. It is deliberately not represented as malicious ground truth.

5,442	32,658	6
CANONICAL OBSERVATIONS	DETECTOR EVALUATIONS	FINAL FINDINGS
all raw-located	5 matched	after aggregation

Measure	Accepted R3 state
Raw evidence artifacts	6
Canonical observations	5,442
Detector evaluations	32,658
Matched detections	5
Hypotheses	6
Findings	6
Narratives	6
Response records	6
Raw embedded	true
Seal verified	true
Canonical observations raw-located	5,442 / 5,442
Same six-source graph component	75 nodes / 74 edges

> The controlled reference case proves the supported workflow and provenance contract. It does not prove that the activity is malicious, that a phishing incident occurred, or that controlled values are malicious IOCs.

Reference case identity

Accepted case ID: case_6bfc43f87590a1f654df3473d9d15c08e17304ed5030d191b4248ee3f4ee2348

Preserving evidence multiplicity without inflating analytical conclusions

Packet processing accounting

9,672	5,412	4,260
R3 PACKET LOCATORS	**NORMALIZED IPV4 TCP/UDP**	**SUPPORTED-PARTIAL ISSUES**
preserved evidence	supported normalization	accounted, not discarded

Parser errors: 0. TraceSOC does not invent protocol names for unidentified EtherTypes simply to improve apparent completeness.

The analytical inflation problem

An earlier form of the Sysmon-to-PCAP correlation produced 66 raw pairwise packet relationships. Those relationships were valid evidence, so deleting them would have violated the evidence-preservation contract. But promoting every packet pair into a separate downstream hypothesis/finding would have inflated the analyst-facing conclusion count.

The repair

- Preserve all 66 raw packet relationships.
- Group them into three source-level analytical hypotheses containing 6, 18 and 42 packet relationships.
- Keep the downstream analyst-facing outputs at six hypotheses/findings/narratives/responses across the complete R3 case.

> This is a key TraceSOC trade-off: preserve evidence multiplicity at the evidence layer, control analytical inflation at the reasoning layer.

Mechanical downstream-to-original-evidence verification

TraceSOC's defining proof is that a material downstream analytical object can be walked backward through every supporting layer to exact raw evidence and original artifact identity.

```
RESPONSE
-> NARRATIVE
-> FINDING
-> HYPOTHESIS
-> UNIQUE MATCHED DETECTION + RELATIONSHIP
-> MATCHED CORRELATION
-> CANONICAL OBSERVATION
-> RAW LOCATOR
-> EMBEDDED RAW ARTIFACT
-> ORIGINAL ARTIFACT SHA-256
```

Authority: evidence/final-proof/r3-end-to-end-reverse-trace-proof.json

Matched detection_result_id values cited by hypotheses are unique in accepted R3. Global uniqueness is not claimed for every no-match detector evaluation.

> Reverse trace proves lineage. It does not independently prove maliciousness or causal intent.

External real evidence, technical freeze and sealed holdout

External real Zeek evidence

A real IoT-23 Zeek source was used before the finalist freeze as an external evidence test. The objective was ingestion/provenance generalization and accounting, not publisher-label accuracy.

Measure	Result
Source rows	23,145
Canonical observations	23,145
Detector evaluations	138,870
Matched detector results	1,641
Findings	0
Exact raw locators	23,145 / 23,145
Source-count difference	0
Tuning to publisher labels	none

> The 1,641 matched detector results are not publisher ground truth and are not presented as accuracy.

Pre-holdout freeze

The finalist technical candidate was frozen at commit 8751a679f36b4983cdd3d2b0b35f44e9a11c4f39 and tree fc420ee3fd1988cf578c0cf309ee3aefcccf33b2 before semantic holdout access. After freeze, product source was not tuned to holdout results.

Sealed post-freeze holdout

Measure	Result
Source rows	156,458
Canonical observations	156,458
Detector evaluations	938,748
Matched detections	0
Findings	0
Raw embedded	true
Seal verified	true
Post-freeze product tuning	none
HOLDOUT_STATUS	CLOSED_FOR_INGESTION_PROVENANCE_GENERALIZATION
DETECTION_ACCURACY_STATUS	UNADJUDICATED

No precision, recall, false-positive or false-negative score is derived because the available publisher-labeled artifact is not an exact valid comparator for the sealed holdout input.

> Zero holdout detections are neither a success claim nor a failure claim. The result closes the measured ingestion/provenance-generalization question and leaves detection accuracy unadjudicated.

Regression, anti-hardcoding, grader closure and deterministic reproduction

Current regression authority

524	524	524
DISCOVERED	RUN	PASSED
current frozen candidate	current frozen candidate	4 / 4 acceptance classes

Historical development denominators remain continuity evidence only and are not presented as current finalist truth.

Anti-hardcoding and grader closure

- Selected case-specific literals checked against production source: 5.
- Production hits: 0.
- R3 linkage failures: 0.
- Narrative observations: 75.
- Narrative observations missing raw locator: 0.
- incident_declared=false.
- response_actions_executed=false.
- Transferable grader lessons satisfied; case-specific historical lessons correctly treated as not applicable where appropriate.

Deterministic clean reproduction

- Two fresh frozen six-source runs were executed.
- Both produced the same 11 core output hashes.
- Those hashes also matched accepted R3.
- Measured runtimes were approximately 3.438 and 3.393 seconds.

> Clean reproduction demonstrates deterministic accepted outputs. It is not a claim of universal detector accuracy.

Measured behavior on the constrained workstation

TraceSOC was developed and validated under a practical resource constraint rather than relying on a heavyweight platform. The accepted scale benchmark records measured behavior on a real-format workload.

Measure	Result
Expected / accepted records	198,463 / 198,463
Quarantined	0
Semantic events	198,463
End-to-end runtime	15.9617856 seconds
Peak working set	486,924,288 bytes
Frozen working-set ceiling	4,000,000,000 bytes
Machine physical RAM	approximately 7.85 GiB

The small-run reproduction record also observed a low child-process working set, but that value is not total system memory and is not used as the main scalability claim.

> The benchmark proves measured resource behavior on the tested workload and machine. It does not establish universal throughput or deployment scale.

Operational and business significance without invented ROI

What becomes clearer

- Evidence identity and exact locators remain available after normalization and correlation.
- Observed fact remains distinguishable from inference, confidence, alternatives and residual uncertainty.
- A reviewer can see what a relationship proves and what it explicitly does not prove.

What becomes safer analytically

- Unsupported or partially supported evidence is not silently dropped.
- Correlation is not silently promoted into causality or maliciousness.
- Controlled values are not promoted into confirmed-malicious IOCs.
- Response recommendations are not represented as executed containment.

What becomes more resilient

- Analyst handoff is less dependent on narrative memory because raw support remains recoverable.
- Accepted outputs can be regenerated from the frozen candidate and accepted inputs.
- Audit and challenge are supported by machine-readable evidence indices and proof artifacts.
- A modest workstation can execute the measured real-format workload without crossing the frozen memory ceiling.

What is not claimed

- No measured monetary savings or ROI.
- No measured analyst-hours saved or incident-response time reduction.
- No false-positive reduction percentage.
- No universal detector accuracy.
- No breach-cost reduction claim.

> The demonstrated business value is investigation assurance: defensible handoff, auditability, exact evidence recovery, repeatability, explicit uncertainty and bounded resource behavior.

What remains unresolved - and why that matters

Limitation / trade-off	Why it exists	Future validation or mitigation
Controlled reference is not malicious ground truth	It is designed to prove interoperability/provenance, not compromise.	Use independent labeled corpora for accuracy questions.
Holdout detection accuracy is UNADJUDICATED	No exact valid labeled comparator exists for the sealed input.	Acquire an exact labeled comparator or a new independently labeled holdout.
Cross-publisher network diversity not established	Development and sealed network corpora are both IoT-23.	Evaluate a genuinely independent publisher/corpus under the same freeze discipline.
No live containment proof	Response records preserve recommendation/validation semantics.	Future controlled deployment with explicit execution authorization and audit.
Explicit representation boundary	Universal parsing would hide semantics and increase unsupported inference risk.	Add representations only with explicit adapter contracts and evidence tests.
Resource results are workload-bounded	One benchmark cannot represent all deployment scales.	Measure additional workloads and machines without changing the accepted claims retroactively.

Why the limitations strengthen the defense

The finalist claim is deliberately narrower than the product could be marketed. That is intentional. The project is built around evidence-preserving analytical discipline; hiding unresolved accuracy or diversity questions would contradict the system's own design principles.

Judge procedure, evidence map and selected authority hashes

Recommended verification order

1. Verify the frozen technical candidate and pre-holdout boundary.
2. Verify current regression/public acceptance against the frozen candidate.
3. Verify controlled R3 counts and 5,442/5,442 raw-locator coverage.
4. Walk the complete reverse-trace proof to original artifact SHA-256.
5. Verify external real Zeek source reconciliation.
6. Verify sealed holdout reconciliation, no post-freeze tuning and unadjudicated accuracy.
7. Verify two fresh reproductions and the same 11 core hashes.
8. Verify the measured scale/resource benchmark.
9. Use evidence-index.csv and finalist-evidence-ledger.json to audit individual report claims and does-not-prove boundaries.

Selected authority hashes

Authority	SHA-256
project-control/finalist-candidate-freeze-v04.json	8cc67c7b17f9a168938fba9119c518f4ac1dd748932419539f5215f0edeceae0
results/acceptance/public-test-results.json	ed379e5d1672569e772df8b3f59fff98991ea298dc41e5ef51e78bae4716230a
evidence/reference-case/case-manifest.json	35e799edea98e91624f4dac8442e75479e50f8cbcc4b21d0a571ad6d6319fdba
evidence/final-proof/r3-end-to-end-reverse-trace-proof.json	cf32024dd0af104a27e8174343483ecc902bd8d4334c724001c80e43e99515db
evidence/final-proof/holdout-final-adjudication.json	cf4764f865a07d75bea647706434fa405b9ac65fb047061cb570511f5785d80b
evidence/final-proof/frozen-clean-reproduction-resource-validation.json	36c2764bc8dbd87a3b567620d9d2958531abdbd7b5cbc7f5f3fd32a881621857
evidence/final-proof/r3-pcap-processing-issue-final-adjudication.json	0a322b28e3b1dafe6fdddb774746267d440db866a96f835c0cf826692f51dadd
evidence/final-proof/final-grader-feedback-closure-audit.json	db54bfea97dc8777296388479d66a164ff7de1e8df98659683f02a38dbd41065
evidence/finalist-problem-impact-proof-defense.md	792b6d093ebdbb88e4cdc9f7de596655d2552ede568e791b327fd89a2109ee2d
finalist-evidence-ledger.json	2712a26b628a3e40d641d83dc2ba5b194cd1c14afad3b84d84047d18bd658ec0
assessment-manifest.json	833dfce9304262c3ae23e2951eb4c25ba05e7dbcb66e7f441c149ef2252a7ade
continuity-record.md	884ae876ef78b7e1a8baab59e205af75fcfe957ffca7eb6554f359a3b3f62ab0

Major technical choices, trade-offs and likely challenge questions

Challenge	Defensible answer
Why provenance-first instead of alert-first?	Because conclusions must survive handoff and challenge. Recoverable evidence makes the analytical result auditable rather than narrative-only.
Why not drop unsupported evidence?	Unsupported does not mean irrelevant. Silent loss creates false completeness and can erase evidence needed for later review.
Why representation-specific adapters?	Different formats have different semantics and locator models. Explicit adapters preserve those differences instead of forcing a misleading universal schema.
Why bounded correlations?	Time adjacency, byte identity and process attribution can corroborate evidence but do not automatically prove user action, causality or malicious intent.
Why aggregate downstream but preserve raw packet relationships?	The evidence layer must retain multiplicity; the reasoning layer must avoid inflating one source-level fact into dozens of analyst-facing conclusions.
Why freeze before holdout?	To separate development from evaluation and prevent unseen evidence from becoming another tuning fixture.
Why is zero holdout detection not a failure or success?	Because accuracy cannot be adjudicated without an exact valid labeled comparator. The measured result is ingestion/provenance behavior, not detector performance.
Why zero confirmed-malicious IOC rows?	Because the reference evidence does not independently establish a malicious IOC. Populating the file anyway would fabricate certainty.
Did TraceSOC execute containment?	No. Response records preserve recommendations/validation but finalist proof does not claim live production execution.
What is the business value?	Investigation assurance: defensible handoff, auditability, evidence recovery, reproducibility, explicit uncertainty and measured viability on a modest workstation.

Things the report deliberately does not say

- The controlled reference is malicious.
- A confirmed phishing incident was established.
- The controlled values are confirmed malicious IOCs.
- Email caused execution.
- Correlation proves causality.
- Process-instance attribution proves maliciousness.
- External IoT-23 detector matches are publisher ground truth.
- Zero holdout detections prove success or failure.
- TraceSOC has universal detector accuracy or cross-publisher generalization.
- TraceSOC delivered measured monetary ROI or executed live containment.

The bounded finalist claim

TraceSOC addresses a real SOC/DFIR assurance problem: heterogeneous evidence can be normalized and analyzed while the reasoning chain becomes progressively harder to audit. The project responds by treating provenance, uncertainty and bounded relationships as technical contracts rather than narrative preferences.

Within the frozen supported representation boundary, the finalist candidate demonstrates a complete controlled six-source reconstruction, full raw-locator coverage for 5,442 canonical observations, bounded cross-source correlations, evidence-preserving aggregation, one complete downstream-to-original-SHA reverse trace, external real-evidence reconciliation, a sealed post-freeze holdout, current 524/524 regression, deterministic clean reproduction and measured resource viability on the constrained workstation.

Equally important, the system does not convert missing evidence into convenient certainty. The controlled case is not called malicious ground truth; holdout detector accuracy remains UNADJUDICATED; cross-publisher diversity remains open; zero confirmed-malicious IOC rows are preserved as the correct evidence state; response records do not masquerade as executed containment; and business value is described without invented ROI.

> Final technical claim: TraceSOC demonstrates evidence-preserving reconstruction, bounded analytical relationships, deterministic reproduction and mechanical downstream-to-raw traceability within its measured supported scope, while keeping unresolved questions explicit instead of manufacturing answers.